Geometry & Topology II: Course Notes

Mathilda Campillo

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Remarks: These are the notes for Geometry and Topology II, Spring 2026. My personal comments will be in teal.

Contents

1 Vector fields on manifolds

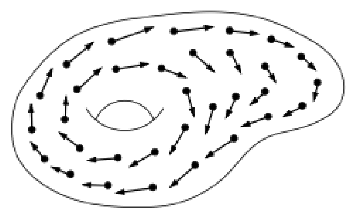
Definition: A vector field on a manifold M (with or without boundary) is a section of the map

$$\pi : TM \to M.$$

More concretely, is a map $X : M \to TM$ with the property that

$$\pi \circ X = \mathrm{Id}_M .$$

We should think about this as in usual calculus, as an arrow attached to each point of M, chosen to be tangent to M, and that vary continuously from point to point.



A vector field

1.1 Local coordinates description

If we suppose that everything is smooth, then given any coordinate chart $(U, (x^i))$ on M^n, we can write the value of X at any point $p \in M$ in terms of the coordinate basis vectors:

$$X_p = X^i(p) \frac{\partial}{\partial x^i}\bigg|_p .$$

This in turn defines n functions $X^i : U \to \mathbb{R}$, called the component function of X in the given coordinate chart.

Examples

1. *Vector field on the plane:* $V : \mathbb{R}^2 \to T\mathbb{R}^2$
$$V(x,y) = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}.$$
2. *Coordinate vector field:* Given a smooth coordinate chart $(U,(x^i))$, the assignment
$$p \mapsto \left.\frac{\partial}{\partial x_i}\right|_p$$
defines a vector field on U, which is smooth because its component functions are constants.
3. *Angle Vector Field on S^1:* Let θ be an angle coordinate on an open subset of S^1. The coordinate vector field $\frac{\partial}{\partial\theta}$ is invariant under changes of angle coordinates (which differ by constants), and thus defines a globally well-defined smooth vector field on S^1. Its component function is constant in any such chart.
4. *Angle Vector Fields on $\mathbb{T}^n$:* On the n-torus $\mathbb{T}^n$, choosing angle coordinates $(\theta^1,\dots,\theta^n)$ gives rise to vector fields
$$\frac{\partial}{\partial\theta^1},\dots,\frac{\partial}{\partial\theta^n},$$
which are smooth and globally defined. These arise from the circle factors and generalize the angle vector field on S^1.

Proposition [Extension Lemma for vector fields]: Let M be a smooth manifold, and let $A \subseteq M$ be closed. Suppose X is a smooth vector field along A. Given any open subset U containing A, there exists a smooth global vector field $\tilde{X}$ on M such that $\tilde{X}|_A = X$ and $\mathrm{supp}(\tilde{X}) \subset U$.

An important remark is that one can extend any vector on a point to be a vector field on the entire manifold! Moreover, if we think about vector fields on submanifolds, then one would get that if S is embedded and X is a vector field along S then there exists a vector field defined on a neighborhood of S in M that agrees with X over S. Furthermore, such vector field extends to all of M is and only if, S is properly embedded.

Let us consider now the assignment $p \mapsto v$ for some $p \in M$ and $v \in T_pM$. Then this defines a vector field along the closed set $\{p\}$. This vector field can be "smoothed out" in a coordinate neighborhood of p by considering a constant-coeff. vector field along such neighborhood. This and the previous proposition yields the following result.

Proposition Let M be a smooth manifold. Let $p \in M$ and $v \in T_pM$. Then, there exists a smooth global vector field X on M such that $X_p = v$.

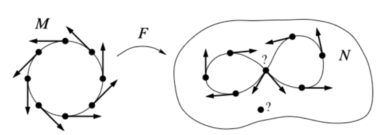
If M is a smooth manifold, then it is common to denote the set of smooth vector fields on M by $\mathfrak{X}(M)$. This set can be endowed with the structure of a vector space with pointwise addition and scalar multiplication. Moreover, vector fields can be multiplied by smooth real-valued functions as follows: Given $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$, then
$$(fX)p = f(p)X_p.$$
This endowes $\mathfrak{X}(M)$ with the structure of module over the ring $C^\infty(M)$.

Exercise

1. Given $X, Y \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$. Prove that $fX + gY \in \mathfrak{X}(M)$.
   A vector field $Z \in \mathfrak{X}(M)$ if and only if, in every local chart (U, φ) on M with coordinates $x^1, \ldots, x^n$, we have
   $$Z\big|_U = \sum_{i=1}^n Z^i \frac{\partial}{\partial x^i}$$
   where each coefficient function $Z^i \in C^\infty(U)$. Let (U, φ) be an arbitrary chart on M. Since $X, Y \in \mathfrak{X}(M)$, we may write
   $$X\big|_U = \sum_{i=1}^n X^i \frac{\partial}{\partial x^i}, \qquad Y\big|_U = \sum_{i=1}^n Y^i \frac{\partial}{\partial x^i}$$
   with $X^i, Y^i \in C^\infty(U)$. The functions $f, g \in C^\infty(M)$ restrict to smooth functions on U (still denoted f and g). Now consider the linear combination:
   $$\begin{aligned}(fX + gY)\big|_U &= f\Big(\sum_{i=1}^n X^i \frac{\partial}{\partial x^i}\Big) + g\Big(\sum_{i=1}^n Y^i \frac{\partial}{\partial x^i}\Big)\\ &= \sum_{i=1}^n \big(fX^i + gY^i\big)\frac{\partial}{\partial x^i}.\end{aligned}$$
   For each i, the coefficient $fX^i + gY^i$ is smooth on U. Thus every local coefficient of $fX + gY$ is smooth. Since the chart (U, φ) was arbitrary, $fX + gY$ is a smooth section of $TM \to M$, i.e.,
   $$fX + gY \in \mathfrak{X}(M).$$

1.2 Pushforward of Vector Fields

Motivation: Given a section of $\sigma : TM \to M$ and a smooth map $F : M \to N$, one gets a map
$$dF : T_pM \to T_{F(p)}N$$
given by
$$dF(\sigma)(p) = dF_p(\sigma_p) \in T_{F(p)}N$$



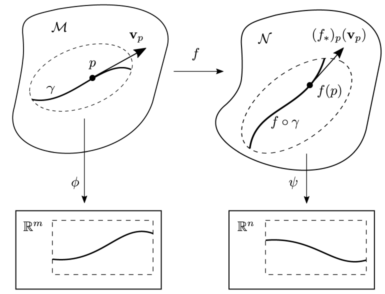
which "naturally pushes" vectors forward. However, this does not in general produce a vector field on N.

Notice that a map going the "opposite" direction is not always possible. Namely, given a vector field on N and F as above, we cannot always pullback this to a vector field on M. Though this is trivially true if F is a diffeomorphism. We will see that the objects that have natural pullbacks are dual to these, and are called differential forms.

Definition: Let $F : M \to N$ be a smooth diffeomorphism, and let X be a smooth vector field on M. Moreover let $p \in N$ and $q := F^{-1}(p) \in M$. Then the <u>pushforward</u> of X by F is the vector field F_*X on N defined by
$$(F_*X)_p \equiv dF_q\big(X_q\big), \qquad p \in N.$$

In words, to compute $F_* X$ at a point $p \in N$, we:

1. pull p back to $F^{-1}(p) \in M$,
2. evaluate X there,
3. then apply the differential dF.



Pushforward of a vector field

1.3 Local and Global frames

Coordinate vector fields on charts provide a convenient way of representing vector fields because their values form a basis of T_pM at each $p \in M$. Clearly, this is not the only model of basis for the tangent space.

Suppose M is a smooth manifold. An ordered k-tuple of vector fields $(X_1, X_2, \ldots, X_k)$ on some subset $A \subseteq M$ is said to be linearly independent if $(X_1|_p, X_2|_p, \ldots, X_k|_p) \in T_pM$ is linearly independent for each $p \in A$, and it is said to span the tangent bundle if at each point $p \in A$ they span T_pM.

Definition: A Local frame for M is an ordered n-tuple of vector fields $(E_1, \ldots, E_n)$ defined on a subspace $U \subseteq M$ such that at each point $p \in U$ they define a basis for T_pM. We say that this frame is global if $U = M$, and we will add the adjective *smooth* if every E_i is smooth.

Notice that we are adding adjectives by considering properties of the k-tuple of vector fields at every point. One can think about imposing that the $E_i|_p$'s are orthonormal with respect to the Euclidean dot product at each point. Then a local (or global) frame that is also orthonormal, will be called a local (or global) orthonormal frame.

As one may think, local frames are plentiful unlike global ones ☹. A smooth n-manifold is said to be *parallelizable* if it admits a smooth global frame. Parallelizability of a manifold is intimately related to wether its tangent bundle can be trivialized.

Some examples: $\mathbb{S}^1$, $\mathbb{S}^3$, $\mathbb{S}^7$, $\mathbb{R}^n$, $\mathbb{T}^n$. Some non-examples: $\mathbb{S}^2$ (Hint: Hairy ball theorem).

1.4 Vector fields as derivations of $C^\infty(M)$

Vector fields define operators on the space of smooth real-valued functions. Notice that if $f \in C^\infty(U)$ and $X \in \mathfrak{X}(M)$. Then we obtain a new function by defining

$$(Xf)(p) \equiv X_p f.$$

Since the action of a tangent vector on a function is locally determined in an arbitrarily small neighborhood. It follows that Xf is locally determined. In particular, for $V \subseteq U$, $(Xf)|_V = X(f|_V)$. This gives us another criterion for smoothness of vector fields.

A smooth vector field $X \in \mathfrak{X}(M)$ defines a map from $C^\infty(M)$ to itself via the assignment $f \mapsto Xf$. This map is clearly $\mathbb{R}$-linear and the product rule for tangent vectors* translates into the following product rule for vector fields:

$$X(fg) = fXg + gXf.$$

A map with such characteristics is called a *global derivation*.

*: Recall that a vector $v \in T_pM$ is a derivation in the following sense: it is a linear map

$$v : C^\infty(M) \to \mathbb{R}$$

satisfying the product rule

$$v(fg) = f(p)\,v(g) + g(p)\,v(f)$$

for each $f, g \in C^\infty(M)$. This defines the product rule for tangent vectors.

1.4.1 Local expression

Let us see how this operator looks in local coordinates. Let M be a smooth manifold, $(U, (x^i))$ a local coordinate chart near $p \in M$, and $f \in C^\infty(M)$. Then $X_p = \sum_{i=1}^n a_i(p) \frac{\partial}{\partial x^i}\Big|_p$ and

$$(Xf)(p) = \sum_{i=1}^n a_i(p) \frac{\partial f(x^1, \dots, x^n)}{\partial x^i}\Bigg|_p$$

in this local chart.

Proposition: Let M be a smooth manifold (with or without boundary). A map $D : C^\infty(M) \to c^\infty(M)$ is a global derivation if, and only if, it is of the form $Df = Xf$ for some smooth vector field $X \in \mathfrak{X}(M)$

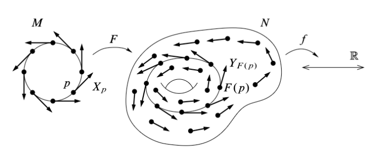
1.5 Vector fields and smooth maps

Resuming a previous discussion about "pushforwards" of vector fields being vector fields, we notice that for example if $F : M \to N$ is not surjective then there is no way to decide which vector to assign to a point not in the image of F, on the other hand, if F is not injective then for some points on N there may be several different vectors obtained by applying dF to some tangent vector on M. We make sure nothing goes wrong after applying the differential by imposing the following:

Definition: Suppose $F : M \to N$ is a smooth map, $X \in \mathfrak{X}(M)$, and $Y \in \mathfrak{X}(M)$. Then, X and Y are said to be <u>F-related</u> if

$$dF_p(X_p) = Y_{F(p)}.$$

The next proposition will enable us to make the idea of a pushforward more precise.



F-related vector fields

Proposition: Suppose M and N are smooth manifolds, and $F : M \to N$ is a diffeomorphism. For every $X \in \mathfrak{X}(M)$, there is a unique smooth vector field on N that is F-related to X.

With this, we define the pushforward of X by F, denoted by F_*X, as the unique vector field on N, F-related to X.

1.6 Lie Brackets

Let X and Y be smooth vector fields on M. Given a smooth function $f : M \to \mathbb{R}$, we can apply X to f to get another smooth function on M. In turn, we can apply Y to the resulting function Xf to obtain yet another smooth function YXf on M. Furthermore, we can do the same in the opposite order to get a priori a (usually) different smooth function on M XYf. Applying both these operations to f and subtracting, we get a well defined operator

$$[X,Y] : \mathbb{C}^\infty(M) \to C^\infty(M)$$

called, the <u>*Lie Bracket*</u> of X and Y, defined by

$$[X,Y]f \equiv XYf - YXf.$$

One can easily check by computation that the Lie bracket of a pair of smooth vector fields defines a global derivation on $C^\infty(M)$ or in other words a smooth vector field.

1.6.1 Coordinate expression for the Lie bracket

Let $X, Y \in \mathfrak{X}(M)$, and let $(U, (x^i))$ be a smooth coordinate chart. Then $[X,Y]$ can be written as

$$[X,Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j} = (XY^j - YX^j) \frac{\partial}{\partial x^j}$$

on that coordinate chart.

1.6.2 Properties of the Lie bracket

The Lie bracket satisfy the following properties for all smooth vector fields $X, Y, Z \in \mathfrak{X}(M)$: (notice that these are the conditions for being a Lie Algebra)

1. Bilinearity
2. Antisymmetry
3. Jacobi Identity:
$$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$$
4. For $f, g \in C^\infty(M)$,
$$[fX, gY] = fg[X,Y] + (fXg)Y - (gYf)X$$

The Lie bracket is natural in the following sense: Given $F : M \to N$, $X_1, X_2 \in \mathfrak{X}(M)$, and $Y_1, Y_2 \in \mathfrak{X}(N)$, such that X_i is F-related to Y_i. Then $[X_1, X_2]$ is F-related to $[Y_1, Y_2]$. Furthermore, $F_*[X_1, X_2] = [F_* X_1, F_* X_2]$

2 Integral curves and Flows

Associated to any smooth vector field there is a primary geometric object called its *integral curves*. Which in words, are smooth curves whose velocity at each point is equal to the value of the vector field there. Moreover, the collection of all integral curves of a given vector field determines a family of diffeomorphisms of (open subsets) of the manifold, called a *flow*.

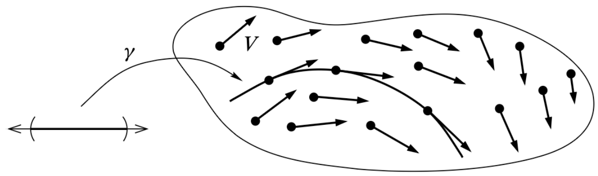


Figure 1: An integral curve of a vector field.

2.1 Integral curves

Suppose M is a smooth manifold. If $\gamma : I \to M$ is a smooth curve, then for each $t \in I$, the velocity vector $\gamma'(t)$ is a vector in $T_{\gamma(t)}M$. We want a curve whose velocity at each point is equal to the value of a given vector field at that point.

Definition: If V is a vector field on M, and integral curve of V is a differentiable curve $\gamma : I \to M$ such that

$$V_{\gamma(t)} = \dot{\gamma}(t).$$

We say γ starts at $p \in M$ if $0 \in I$ and $\gamma(0) = p$.

Given a smooth vector field on a manifold and point on it, one can always find an integral curve defined on a small interval. That is,

Proposition: Let $V \in \mathfrak{X}(M)$. For each point in $p \in M$, there exists $\epsilon > 0$ and a smooth curve $\gamma : (-\epsilon, \epsilon) \to M$ such that γ is an integral curve of V starting at p.

Remark: A natural question to ask is: how affine reparametrizations affect integral curves? The answer is: If you consider a reparametrization that rescales, then you will get an integral curve of the rescaled vector field, and if you translate then you will get an integral curve of the same vector field.

2.1.1 Naturality of integral curves

Suppose M and N are smooth manifolds and $F : M \to N$ is a smooth map. Then, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are F-related if and only if F takes integral curves of X to integral curves of Y, meaning that for each γ an integral curve of X, $F \circ \gamma$ is an integral curve of Y.

If in particular, F is a smooth diffeomorphism then $F \circ \gamma$ defines an integral curve of the pushforward of X along F.

2.2 Flows

Motivation: There is another useful way to visualize the family of integral curves associated to a vector field. Let M be a smooth manifold and let $V \in \mathfrak{X}(M)$ be a smooth vector field. Suppose that for every point $p \in M$, there exists a unique integral curve of V starting at p and defined for all $t \in \mathbb{R}$. We denote this curve by

$$\gamma_p : \mathbb{R} \to M.$$

(Although in general integral curves need not be defined for all t, we assume this here for simplicity.)

For each $t \in \mathbb{R}$, define a map

$$\varphi_t : M \to M$$

by

$$\varphi_t(p) = \gamma_p(t).$$

In other words, φ_t moves each point p along its integral curve for time t. By the translation property of integral curves, the curve $t \mapsto \gamma_p(t+s)$ is an integral curve starting at the point $q = \gamma_p(s)$. By uniqueness, this implies

$$\gamma_q(t) = \gamma_p(t+s).$$

Rewriting this in terms of the maps φ_t, we obtain

$$\varphi_t(\varphi_s(p)) = \varphi_{t+s}(p).$$

Additionally, since $\gamma_p(0) = p$, we have

$$\varphi_0(p) = p.$$

Therefore, the map

$$\varphi : \mathbb{R} \times M \to M, \quad (t,p) \mapsto \varphi_t(p)$$

defines an action of the additive group $\mathbb{R}$ on M.

Definition: Let V be a smooth vector field on a smooth manifold M. Let $t \geq 0$, and let

$$U_t \subset M$$

be the set of points p for which there exists an integral curve $\gamma : [0,t] \to M$ passing through p. We define the *time-t flow*

$$\varphi_t^V : U_t \to M$$

to be the map sending each $p \in U_t$ to $\gamma(t)$, where γ is the integral curve with $\gamma(0) = p$. We define the time $-t$ flow to be the time t flow of $-V$. The family of maps $(\varphi_t)_{t\in\mathbb{R}}$ is called the *flow* of V. Finally, we say that V is *complete* if the domain of φ_t is M for all $t \in \mathbb{R}$.

Note that we have $U_t = M$ for all t if M is closed. In this case, the time-t flow is a map $\varphi_t^V : M \to M$.

Example

1. Consider the vector field $V := \frac{\partial}{\partial x}$ on $\mathbb{R}^2$ with coordinates (x, y). Define
$$\varphi : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2, \quad \varphi(t, (x, y)) := (x + t, y),$$
and
$$\varphi_t(x, y) := \varphi(t, (x, y)).$$
Then the path
$$\gamma_{(x,y)} : \mathbb{R} \to \mathbb{R}^2, \quad \gamma_{(x,y)}(t) := \varphi(t, (x, y))$$
is a flow line of V starting at (x, y). Hence $(\varphi_t)_{t\in\mathbb{R}}$ is the flow of V.
If we replace $\mathbb{R}^2$ with a smaller open subset (for example, an open ball $U \subset \mathbb{R}^2$), then the time-3 flow may not exist, since the flow line can leave the domain.
2. Consider the vector field
$$V := x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}$$
on $\mathbb{R}^2$. Define
$$\varphi : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2, \quad \varphi(t, (x, y)) := (\cos t\, x, \sin t\, y),$$
and
$$\varphi_t(x, y) := \varphi(t, (x, y)).$$
Then
$$\gamma_{(x,y)}(t) := \varphi(t, (x, y))$$
is a flow line of V, and $(\varphi_t)_{t\in\mathbb{R}}$ is the flow of V.

Definition: A <u>one-parameter subgroup</u> of a group G is a group homomorphism $\mathbb{R} \to G$.

This tells us that the map $t \to \varphi_t^V$ is a one-parameter subgroup of $Diff(M)$

Definition: A <u>smooth one-parameter family of diffeomorphisms</u> of M is a family of diffeomorphisms $(\varphi_t)_{t\in\mathbb{R}}$ such that
$$\varphi : \mathbb{R} \times M \to M \quad \varphi(t, p) \equiv \varphi_t(p)$$
is a smooth map. A <u>smooth one-prameter subgroup of diffeomorphisms</u> of M is a smooth family of diffeomorphisms as above such that
$$\mathbb{R} \to Diff(M) \quad t \to \varphi_t$$
is a one-parameter subgroup.

For a smooth one-parameter family of diffeomorphisms $(\varphi_t)_{t\in\mathbb{R}}$ as above we define $\frac{\partial\varphi}{\partial t}(\tau)$ to be the smooth vector field whose value at $p \in M$ is
$$D\varphi\left(\frac{\partial}{\partial t}\right)\bigg|_{(\tau,\tau^{-1}(p)}$$

Proposition: Let $\varphi : \mathbb{R} \times M \to M$ be a smooth map so that $\varphi_t \equiv \varphi|_{t\times M} : M \to M$ is a smooth diffeomorphism for each $t \in \mathbb{R}$, and so that $\varphi_{t+s} = \varphi_t \circ \varphi_s$ for each $s, t \in \mathbb{R}$. Then, there exists a smooth vector field $V \equiv \frac{\partial\varphi}{\partial t}()$ on

M, whose value at a point $p \in M$ is $D\varphi\left(\frac{\partial}{\partial t}\right)\Big|_{(0,p)}$, and whose flow is given by the family $(\varphi_t)_{t\in\mathbb{R}}$. Moreover, the assignment $p \mapsto V_p$ is a (rough) vector field called the infinitesimal generator of φ.

Proposition: Let $F : M \to N$ be a smooth diffeomorphism, and let $(\varphi_t)_{t\in\mathbb{R}}$ be the flow of a smooth vector field V on M. Define $\phi_t = F \circ \varphi_t \circ F^{-1}$, then $(\phi_t)_{t\in\mathbb{R}}$ is the flow of F_*V.

Every smooth global flow gives rise to a smooth vector field whose integral curves are precisely the curves defined by the flow. Conversely, we would now like to say that every smooth vector field is the infinitesimal generator of a smooth global flow. However, it is easy to see that one cannot obtain such result since there are vector fields whose flows are not defined for all time.

Definition: We say a vector field is complete if it generates a global flow, or equivalently, if each of its integral curves are defined for all $t \in \mathbb{R}$.

Proposition: Every compactly supported smooth vector field on a smooth manifold is complete.

Proposition: On a compact smooth manifold, every smooth vector field is complete.

2.3 Lie Derivatives

Motivation: Let M and N be smooth manifolds of dimensions m and n, respectively. For any smooth map

$$F : M \longrightarrow N,$$

we can naturally associate a bundle map

$$dF : TM \longrightarrow TN,$$

often called the *global differential* of F. This construction generalizes the familiar differential of a smooth map

$$G : \mathbb{R}^m \longrightarrow \mathbb{R}^n,$$

and can be regarded as the derivative of F in the context of manifolds. The global differential is a standard concept that appears early in most introductory courses on differential geometry.

2.3.1 Why does the naive differential fail?

To understand the Lie derivative, we must first recognize why standard calculus cannot be directly applied to vector fields on manifolds.

The Global Differential Problem: Given a vector field Y on a manifold M, we can view it as a map $Y : M \to TM$. Its differential is $dY_p : T_pM \to T_{Y_p}(TM)$. This results in a vector in the tangent space of the tangent bundle, which is a $2n$-dimensional space. We want a derivative that stays within the n-dimensional tangent space T_pM.

The Projection Obstruction: If we try to project this differential back down to M using the natural projection $\pi : TM \to M$, we find that $\pi \circ Y = \mathrm{id}_M$. By the chain rule:

$$d\pi_{Y_p} \circ dY_p = d(\mathrm{id}_M)_p = \mathrm{id}_{T_pM}.$$

Sophus Lie in 1896

This means the "horizontal" part of the differential dY always yields the identity map, regardless of how the vector field Y actually behaves. It fails to capture the "change" in the field.

Disjointness of Tangent Spaces: The fundamental issue is that T_pM and T_qM are disjoint for $p \neq q$. In $\mathbb{R}^n$, we can subtract vectors at different points because there is a canonical way to identify all tangent spaces. On a manifold, there is no natural way to compare a vector at p with a vector at q without additional structure (like a flow or a connection).

2.3.2 Good way to do it! Geometric definition via flows

The Lie derivative $\mathcal{L}_X Y$ solves the comparison problem by using the **flow** θ_t of a vector field X to "transport" the vectors back to a single point.

Definition: Let θ_t be the flow of X. The Lie derivative of Y along X at point p is:

$$\mathcal{L}_X Y|_p = \lim_{t \to 0} \frac{(d\theta_{-t})_{\theta_t(p)}\big(Y_{\theta_t(p)}\big) - Y_p}{t}.$$

Basically we are "dragging" the vector field Y back from the point $\theta_t(p)$ to the point p using the differential of the flow, then perform the standard subtraction in T_pM.

2.3.3 The Lie bracket in local coordinates

The Lie derivative is equivalent to the Lie bracket $[X, Y]$. In local coordinates x^i, if $X = X^i \partial_i$ and $Y = Y^j \partial_j$, the components are:

$$[X, Y]^j = X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i}.$$

This can be interpreted as the commutator of the two vector fields acting as derivations on smooth functions:

$$[X, Y]f = X(Yf) - Y(Xf).$$

2.3.4 Properties of the Lie derivative

For each triple of smooth vector fields $X, Y, Z \in \mathfrak{X}(M)$, the following holds:

1. $\mathcal{L}_X Y = -\mathcal{L}_Y X$
2. $\mathcal{L}_X [Y, Z] = [\mathcal{L}_X Y, Z] + [Y, \mathcal{L}_X Z]$
3. $\mathcal{L}_{[X,Y]} Z = \mathcal{L}_X \mathcal{L}_Y Z - \mathcal{L}_Y \mathcal{L}_X Z$
4. If $g \in C^\infty(M)$, then $\mathcal{L}_X (gY) = (Xg)Y + g\mathcal{L}_X Y$
5. If $F : M \to N$ is a diffeomorphism, then $F_*(\mathcal{L}_X Y) = \mathcal{L}_{F_* X} F_* Y$

2.3.5 The Flow-Commutator Identity

A central result is the relationship between the Lie bracket and the commutator of the flows θ_t (for X) and ϕ_t (for Y).

Consider the curve

$$c(t) = (\phi_{-t} \circ \theta_{-t} \circ \phi_t \circ \theta_t)(p).$$

- The first derivative $c'(0)$ vanishes, which is why the second derivative acts as a derivation.
- The second derivative at $t = 0$ recovers the Lie bracket:

$$\left.\frac{d^2}{dt^2}\right|_{t=0} (\phi_{-t} \circ \theta_{-t} \circ \phi_t \circ \theta_t)(p) = 2[X, Y]_p.$$

Another way to think about this: The Lie derivative $\mathcal{L}_X Y$ can be formally understood as a second-order measure of the failure of the flows θ_t (associated with X) and ϕ_t (associated with Y) to commute. This relationship is captured by the *commutator curve*

$$c(t) = (\phi_{-t} \circ \theta_{-t} \circ \phi_t \circ \theta_t)(p).$$

Because the first-order Taylor terms of the respective flows cancel out, the first derivative $c'(0)$ vanishes. This uniquely allows the second derivative $c''(0)$ to act as a derivation (i.e., a tangent vector) on the manifold, rather than as a higher-order acceleration term. A second-order expansion reveals that the non-vanishing components of this "loop" are precisely the components of the Lie bracket, leading to the identity

$$c''(0) = 2[X, Y] = 2\mathcal{L}_X Y,$$

which interprets the Lie derivative as the infinitesimal displacement resulting from traversing a "square" defined by the two vector fields.

3 Vector bundles

3.1 Cotangent bundle

Definition: The cotangent bundle is defined as

$$T^*M \equiv (TM)^*.$$

We call elements of this dual space, *covectors.* What we are doing here is just taking the dual vector space to T_pM at each point $p \in M$. As in the case of the tangent bundle, smooth local coordinates for M yield smooth local coordinates for its cotangent bundle

Recall: There is a one-to-one correspondence between between tangent vectors at p and derivations at p.

$$\{\text{vectors} \in T_pM\} \longleftrightarrow \{v : C^\infty(M) \to \mathbb{R} : v(fg)|_p = f|_p v(g) + g|_p v(f)\}$$

Notice that given a point $p \in M$, a covector $\omega \in T_p^*M$, and a tangent vector $v \in T_pM$, then one can define a pairing $\langle \omega, v\rangle \in \mathbb{R}$. This follows basically from the definition of dual vector space. Now, given a function f locally defined near $p \in M$ and a tangent vector $v \in T_pM$, one can similarly define a pairing given by

$\langle df_p, v\rangle = v(f)_p \in \mathbb{R}$, where v is considered as a derivation defined near $p \in M$. We now give the precise definition of the differential.

Definition: Let $f : U \to \mathbb{R}$ be a smooth function defined on an open neighborhood of $p \in M$. Then the differential (of f at $p \in M$) is defined as

$$\begin{aligned} df_p : T_pM &\to T_p^*M \\ v &\mapsto df_p(v) \equiv v(f)_p \end{aligned}$$

The differential assigns a scalar to each tangent vector via the pairing in the previous comment.

3.1.1 How does df looks more concretely?

We can compute the differential in local coordinates as follows:

Given a local chart (U, x) near p, then each coordinate function x^i is smooth and has differential $dx^i|_p$. Moreover, the differentials $\{dx^1|_p, \ldots, dx^n|_p\} \subset T_p^*M$ provide a dual basis to $\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\} \subset T_pM$. In these local coordinates, we can write the differential as

$$df_p = \sum_i^n \frac{\partial f}{\partial x^i}\Big|_p dx^i\Big|_p$$

Notice that we can think of df as an analogue of the usual gradient, in a way that makes coordinate-independent sense on a manifold.

Lemma: $dx^i|_p, \ldots, dx_n|_p$ is a basis for T_p^*M.

Some useful remarks:

1. Any covector $\omega_p \in T_p^*M$ can be written uniquely as

$$\omega_p = \omega_i dx^i\Big|_p, \quad \text{where } \omega_i = \omega\left(\frac{\partial}{\partial x^i}\Big|_p\right).$$

   This secretly says that all covectors come from differentials.

2. For each smooth function $f : M \to \mathbb{R}$, the map

$$\begin{aligned} df : M &\to T^*M \\ p &\mapsto df_p \in T_p^*M \end{aligned}$$

   is a smooth section. Why? because at each point we are choosing the linear functional $df_p : T_pM \to \mathbb{R}$. Now, we will later see that these sections will define something called covector fields or differential 1-forms.

Properties: Let $f, g \in C^\infty(M)$. Then

1. $d(af + bg) = adf + bdg$
2. $d(fg) = gdf + fdg$
3. $F : N \to M$ smooth map between smooth manifolds,

$$d(f \circ F)(V) = df(DF(v)) \forall v \in T_qN, q \in N.$$

Slightly off topic sidenote: We know that geometrically, we can think of a vector field on M as a rule that assigns an arrow to each point of M, but what kind of geometric picture can we form of a covector field?

The take away idea is that a nonzero linear functional $\omega_p \in T_p^* M$ is completely determined by two things:

1. Its kernel, which is a codimension one linear subspace of $T_p M$
2. The set of vectors in $T_p M$ such that $\omega_p(v) = 1$, which determines an affine hyperplane parallel to its kernel.

Then, we can visualize a covector field as a defining pair of hyperplanes in each tangent space, one through the origin and another parallel to it, varying continuously from point to point.

Notice that up to this point, we have defined two different kinds of derivatives of f at a point $p \in M$:

1. The pushforward $f_* : T_p M \to T_{f(p)}\mathbb{R}$.
2. The differential $df_p : T_p M \to \mathbb{R}$.

But these are the same locally, once we take into account the canonical identification between $\mathbb{R}$ and its tangent space at any point.

Lemma: $f \in C^\infty(M)$, $f : M \to \mathbb{R}$ smooth, $\gamma : [0,1] \to M$ smooth path. Then

$$F(\gamma(1)) - F(\gamma(0)) = \int_0^1 df(\gamma'(t))dt.$$

Definition: A 1-form is a smooth section ω of T^*M. If $F : N \to M$ is a smooth function, then the pullback of ω is $F^*\omega$ on N, satisfying

$$F^*\omega(v) = \omega(DF(v)), \ \ \forall v \in T_p M.$$

Definition: $\gamma : [a,b] \to M$ be a smooth map. The line integral of ω along γ is

$$\int_a^b \omega(\gamma'(t))dt.$$

Examples: (1) $M = \mathbb{R}^2$ and consider $\omega = xdy$ which is not exact (not even close). Take the path $\gamma : [0,\pi] \to \mathbb{R}^2$ given by $\gamma(t) = (\cos(t), sin(t)$. Then

$$\gamma'(t) = \sin(t)\frac{\partial}{\partial x} + \cos(t)\frac{\partial}{\partial y}$$

$$\omega(\gamma'(t)) = \cos^2(t).$$

$$\int_0^1 \omega(\gamma'(t))dt = \int_0^1 \cos^2(t)dt > 0$$

Now take the path $\Gamma : [0,\pi] \to \mathbb{R}^2$ defined by $\Gamma(t) = (\frac{t}{\pi}, 0)$

4 Tensors

Definition: Let $k, l \in \mathbb{N}$. The (k,l)-tensor bundle on M is defined by

$$T^{(k,l)}(M) = \underbrace{TM \otimes \cdots \otimes TM}_{k \text{ times}} \otimes \underbrace{T^*M \otimes \cdots \otimes T^*M}_{l \text{ times}}.$$

We call k the *contravariant degree* and l the *covariant degree.*

If $k = l = 0$, then

$$T^{(0,0)}(M) \cong M,$$

that is, smooth functions.

Contravariant components come from the tangent bundle and covariant components come from the cotangent bundle. The ordering convention is that contravariant indices appear first.

Examples:

1. $T^{(1,0)}(M) = TM$.
2. $T^{(0,1)}(M) = T^*M$.
3. $$T^{(1,1)}(M) = TM \otimes T^*M \cong \mathrm{Hom}(TM, TM).$$

   So $(1,1)$-tensors can be viewed as linear endomorphisms of each tangent space.

Definition: A (k,l)-tensor field is a smooth section of $T^{(k,l)}(M)$.

4.1 Local coordinates description

Let (U, x) be a local chart with coordinates $(x^1, \ldots, x^n)$ and let $p \in U$. Then

$$\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \ldots, \frac{\partial}{\partial x^n}\bigg|_p \right\}$$

is a basis of T_pM, and

$$\left\{ dx^1\Big|_p, \ldots, dx^n\Big|_p \right\}$$

is the dual basis of T_p^*M.

A general element of $T_p^{(k,l)}M$ can be written as

$$\sum A^{i_1,\ldots,i_k}_{j_1,\ldots,j_l}(p) \frac{\partial}{\partial x^{i_1}} \otimes \cdots \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_l}.$$

The coefficients transform in a specific way under coordinate changes, this is what makes tensors coordinate-independent geometric objects.

4.2 Symmetric and antisymmetric tensors

Definition: Let $\alpha \in T^{(0,k)}(M)$. We say α is

1. Symmetric if
$$\alpha(v_1, \ldots, v_k) = \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$
for all permutations σ.
2. Antisymmetric if
$$\alpha(v_1, \ldots, v_k) = (-1)^{\text{sign}(\sigma)} \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(k)}).$$

Antisymmetric tensors are the ones that will give rise to differential forms.

4.3 Pullback of covariant tensors

Let $F : M \to N$ be smooth and let $\alpha \in \Gamma(T^{(0,k)}(N))$.

Definition: The pullback $F^*\alpha \in \Gamma(T^{(0,k)}(M))$ is defined by

$$(F^*\alpha)(v_1, \ldots, v_k) = \alpha(DF(v_1), \ldots, DF(v_k)).$$

Notice that this only naturally works for covariant tensors. The differential pushes vectors forward, and the pullback compensates by feeding them into α.

5 Differential forms

Definition: A differential k-form on M is a smooth section of

$$\wedge^k(T^*M).$$

We denote the space of all smooth k-forms by $\Omega^k(M)$.

We also define
$$\Omega^*(M) = \bigoplus_{k=0}^{n} \Omega^k(M).$$

A k-form is simply an antisymmetric covariant k-tensor field.

5.1 Local expression

In local coordinates $(x^1, \ldots, x^n)$, a k-form can be written as

$$\alpha = \sum A_{i_1,\ldots,i_k}(p) dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

The wedge product enforces antisymmetry. In particular, $dx^i \wedge dx^i = 0$.

If $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^l(M)$, then
$$\alpha \wedge \beta \in \Omega^{k+l}(M).$$

5.2 Pullback of forms

If $F : M \to N$ is smooth and $\alpha \in \Omega^k(N)$, then

$$(F^*\alpha)(v_1, \dots, v_k) = \alpha(DF(v_1), \dots, DF(v_k)).$$

Pullback preserves wedge products and linearity.

In local coordinates, if

$$F(x^1, \dots, x^m) = (F^1(x), \dots, F^n(x)),$$

then

$$F^*(dy^i) = d(F^i) = \sum_{j=1}^{m} \frac{\partial F^i}{\partial x^j} dx^j.$$

Example:

1. Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be given by

$$F(x, y) = (F^1(x, y), F^2(x, y)).$$

Then

$$F^*(dx \wedge dy) = dF^1 \wedge dF^2.$$

Now compute

$$dF^1 = \frac{\partial F^1}{\partial x} dx + \frac{\partial F^1}{\partial y} dy, \qquad dF^2 = \frac{\partial F^2}{\partial x} dx + \frac{\partial F^2}{\partial y} dy.$$

Therefore,

$$\begin{aligned} dF^1 \wedge dF^2 &= \left(\frac{\partial F^1}{\partial x} dx + \frac{\partial F^1}{\partial y} dy \right) \wedge \left(\frac{\partial F^2}{\partial x} dx + \frac{\partial F^2}{\partial y} dy \right) \\ &= \frac{\partial F^1}{\partial x} \frac{\partial F^2}{\partial x} \, dx \wedge dx + \frac{\partial F^1}{\partial x} \frac{\partial F^2}{\partial y} \, dx \wedge dy \\ &\quad + \frac{\partial F^1}{\partial y} \frac{\partial F^2}{\partial x} \, dy \wedge dx + \frac{\partial F^1}{\partial y} \frac{\partial F^2}{\partial y} \, dy \wedge dy. \end{aligned}$$

Since $dx \wedge dx = 0$ and $dy \wedge dy = 0$, and since

$$dy \wedge dx = -dx \wedge dy,$$

we obtain

$$dF^1 \wedge dF^2 = \left(\frac{\partial F^1}{\partial x} \frac{\partial F^2}{\partial y} - \frac{\partial F^1}{\partial y} \frac{\partial F^2}{\partial x} \right) dx \wedge dy.$$

Thus,

$$F^*(dx \wedge dy) = \det(JF) \, dx \wedge dy.$$

The antisymmetry of the wedge product is exactly what produces the determinant. This is not a coincidence, determinants measure oriented area scaling, and top-degree forms encode orientation. This explains why the Jacobian determinant appears in change of variables for integrals (it is a fundamental statement about how top-degree forms transform).

5.3 Exterior derivatives

Recall: Given $f \in C^\infty(M) = \Omega^0(M)$ $f : M \to \mathbb{R}$ you can take the differential $df \in \Omega^1(M)$.

The goal is to consider an arbitrary p-form and define its associated "differential" that gives you a $p+1$-form.

Theorem: There exists a unique linear map

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

satisfying the following:

1. For each $f \in C^\infty(M)$, $df(v) = v(f)$
2. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$

Lemma: For a smooth map $F : M \to N$, $\alpha \in \Omega^k(N)$. Then

$$F^*(d\alpha) = d(F^*\alpha)$$

A general k-form looks like $fx + gdy + hdz$ in $\mathbb{R}^3$.

Lemma: For every $\alpha \in \Omega^k(M)$, $d(d\alpha) = 0$, i.e. $d^2 = 0$.

The previous lemma helps us to prove uniqueness of the operator d.

Example: $\alpha = f(x,y,z)dx + g(x,y,z)dy + h(x,y,z)dz$. Then

$$\begin{aligned} d\alpha &= df \wedge dx + dg \wedge dy + dh \wedge dz \\ &= (f_y dy + f_z dz) \wedge dx + (g_x dx + g_z dz) \wedge dy + (h_x dx + h_y dy) \wedge dz = \textbf{curl} \end{aligned}$$

5.4 Interior derivative (inner derivation, contraction)

Theorem: Let V be a smooth vector field $V \in \Gamma(M)$ smooth vector field on M. Then there exists a unique linear map $\Omega^k(M) \to \Omega^{k-1}(M)$:

1. $V \lrcorner df = V(f)$
2. $V \lrcorner (\alpha \wedge \beta) = (V \lrcorner \alpha) \wedge \beta + (-1)^k \alpha (V \lrcorner \beta)$ for all $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^(M)$.

Note: $\alpha(V_1, V_2, \ldots, V_k) = (V_k \lrcorner \ldots (V_2 \lrcorner (V_1 \lrcorner \alpha)))$

Proposition: $\omega \in \Omega^1(M)$, $X, Y \in \Gamma(TM)$

$$d\omega(X,Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]).$$

6 Distributions and foliations

def smooth distribution.

Lemma: For all p in M, there exists a neighborhood $p \in U$ and smooth vector fields $X_1, \ldots, X_k$ such that $X_1|_q, \ldots, X_k|q$ is a basis for $D|_q$ for all $q \in U$. Conversely any subset $D \subset TM$ with this property is a rank k distribution.

Def: Given a rank k distribution $D \subset TM$ and integral submanifold of D is a submanifold $N \subset M$ st $T_pN = D|_p$ for all $p \in N$.

Examples

1. V nowhere vanishing vector field defines a rank one distribution $D_p = \text{span}(V(p))$ and the integral submanifold is integral curve or flow line.
2. $\{\partial/\partial x_1\}_1^k$ span a rank k distribution D in $\mathbb{R}^n$ with integral submanifold $\mathbb{R}^k$
3. $D = \text{Span}(X, Y)$, $X = \partial/\partial x + y\partial/\partial z$, $Y = \partial/\partial y$

Def: Involutive distribution

Def: A distribution $D \subset TM$ is integrable if each point $p \in M$ is contained in an integral submanifold of M

Prop: Every integral distribution is involutive.

Ex: Assume there is an integral submanifold $N \subset \mathbb{R}^3$

Lemma: A subset $D \subset TM$ is a rank k distribution iff forall $p \in M$ there exists $p \in U \subset M$ and $\omega_1, \dots, \omega_{n-k} \in \Omega^1(M)$ s.t. for all $q \in U$

$$D|_q = ker(\omega_1)|_q \cap \cdots \cap ker(\omega_{n-k}|_q$$

remark: the omegas are called defining 1-forms of the distribution.

def: A p-form ω annihilates D if for every open subset $U \subset M$ and for all $X_1, \dots, X_p \in \Gamma(D|_U)$

$$\omega(X_1, \dots, X_p) = 0$$

Lemma: Let $\omega_1, \dots, \omega_{n-k}$ bet the defining 1 forms for a given rank k distribution D. A p form $\eta \in \Omega^p(M)$ annihilates D iff there exists $\beta_1, \dots, \beta_{n-k} \in \Omega^{p-1}(M)$ such that

$$\eta = \sum_{i=1}^{n-k} \omega_i \wedge \beta_i$$

Theorem: $D \subset TM$ distribution of codimension 1. Then D is involutive iff for every $U \subset M$ and any annihilating form $\eta \in \Omega^1(D|_U)$ we have that $d\eta$ also annihilates $D|_U$.

def: M , $D \subset TM$ rank k distribution. Then D is involutive if for all $U \subset M$ and for all $X, Y \in \Gamma(D|_U)$, $[X, Y] \in \Gamma(D|_U)$

Prop: $\Gamma(D) \subset \Gamma(TM)$. D is involutive iff $\Gamma(D)$ is a Lie subalgebra of $\Gamma(D)$

Def: Let M, $D \subset TM$ is a rank k distribution. A local chart (U, φ) and local coordinates $(x_1, \dots, x_n)$ is flat for D if $\{\partial/\partial x_j\}$ is a frame for $D|_U$. D is completely integrable if $\forall p \in M$, p is contained in a chart (U, φ) which is flat for D.

Remark: D is completely integrable $\{x_{n-k} = c_{n-k}, \dots, x_n = c_n\}$ is a k dimensional integral submanifold of M.

Theorem Frobenius: Every involutive distribution is completely integrable.

Lemma: Let D be a rank k distribution. Suppose that $\forall p \in M$, there exists $U \in M$ and aa frame $X_1, \dots, X_k$ of $D|_U$ such that $[X_i, X_j] = 0$. Then D is completely integrable.